

Beam Instrumentation and Diagnostics for Particle Accelerators

(Practicas)

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We want to measure the LINAC emittance → The area of the x - x' phase space.

We also know that the emittance depends on the accelerator optics defined by Twiss Parameters

→ The beam matrix Σ **in a given point** of the machine can be expressed both in position-divergence distribution and as a combination of the Twiss Parameters:

$$\Sigma = \begin{pmatrix} \langle x^2 \rangle & \langle xx' \rangle \\ \langle xx' \rangle & \langle x'^2 \rangle \end{pmatrix} = \begin{pmatrix} \sigma_x^2 & \sigma_x \sigma'_x \\ \sigma_x \sigma'_x & \sigma'^2_x \end{pmatrix} = \varepsilon \begin{pmatrix} \beta & -\alpha \\ -\alpha & \gamma \end{pmatrix}$$

To obtain the evolution of position and distribution, the beam matrix is transported using the **transport matrix R** :

$$\Sigma = R \Sigma_0 R^T$$

The only thing we can directly measure is σ

$$\begin{pmatrix} \sigma_x^2 & \sigma_x \sigma'_x \\ \sigma_x \sigma'_x & \sigma'^2_x \end{pmatrix} = \varepsilon R \begin{pmatrix} \beta_0 & -\alpha_0 \\ -\alpha_0 & \gamma_0 \end{pmatrix} R^T$$

R depends on the transport line
→ Known

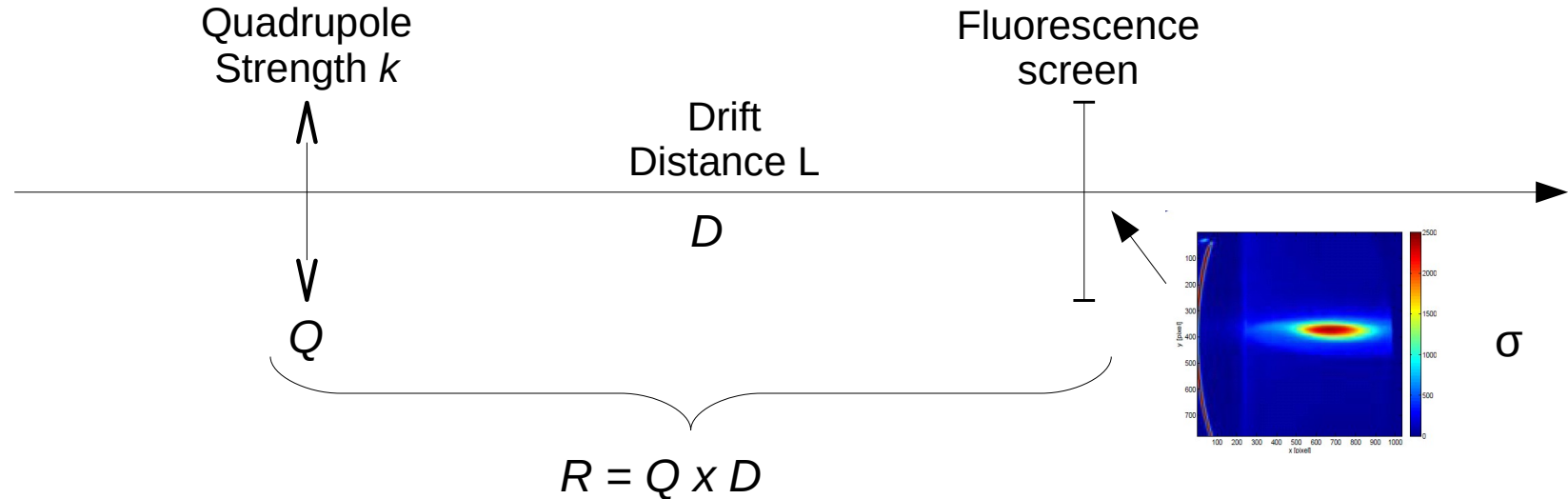
$\alpha_0, \beta_0, \gamma_0, \varepsilon$
are unknown
but not independent:
 $\beta\gamma - \alpha^2 = 1$

$$\sigma_x = f(\alpha, \beta, \gamma, \varepsilon, R_{ij})$$

3 unknown → 3 measurements for different R s

Transport Matrix – Thin Lens

The transport matrix depends on the elements of the lattices in between the two interested longitudinal positions



Quadrupole matrix
(Thin lens approximation):

$$Q = \begin{pmatrix} 1 & 0 \\ -k & 1 \end{pmatrix}$$

Drift matrix

$$D = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \longrightarrow Q \times D = \begin{pmatrix} 1 - kL & L \\ -k & 1 \end{pmatrix}$$

R can be modified by changing the quadrupole strength k

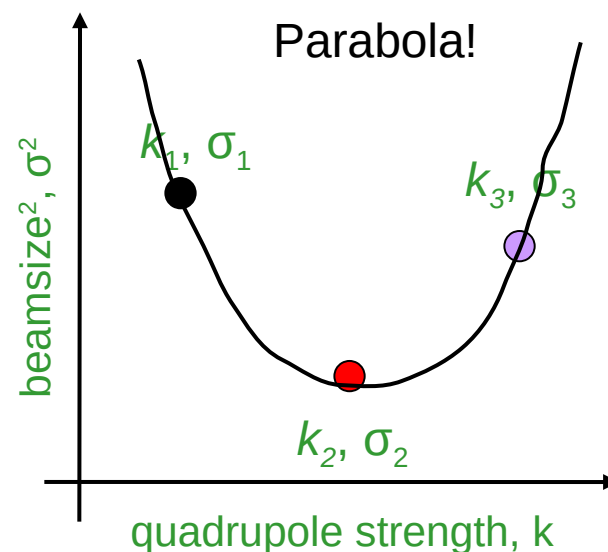
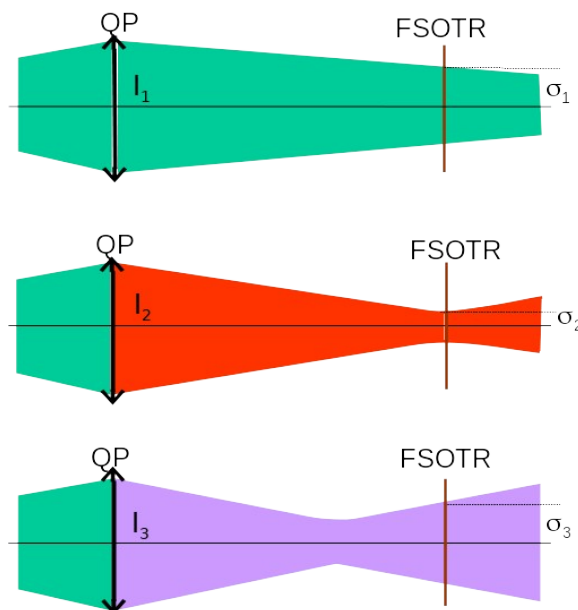
$$\begin{pmatrix} \sigma_x^2 & \sigma_x \sigma'_x \\ \sigma_x \sigma'_x & \sigma_x'^2 \end{pmatrix} = \varepsilon R \begin{pmatrix} \beta_0 & -\alpha_0 \\ -\alpha_0 & \gamma_0 \end{pmatrix} R^T$$

$$\begin{pmatrix} \sigma_x^2 & \sigma_x \sigma'_x \\ \sigma_x \sigma'_x & \sigma_x'^2 \end{pmatrix} = \varepsilon \begin{pmatrix} 1 - kL & L \\ -k & 1 \end{pmatrix} \begin{pmatrix} \beta_0 & -\alpha_0 \\ -\alpha_0 & \gamma_0 \end{pmatrix} \begin{pmatrix} 1 - kL & -k \\ L & 1 \end{pmatrix}$$

→ $\sigma_x^2 = \varepsilon (\beta_0(1 - kL)^2 - 2\alpha_0(1 - kL) - \gamma_0 L^2)$

$$= \varepsilon \left(\beta_0(1 - kL)^2 - 2\alpha_0(1 - kL) - \frac{1 + \alpha_0^2}{\beta_0} L^2 \right) = Ak^2 - 2ABk + C + AB^2$$

I can obtain different beam size when changing the quadrupole strength
→ I am focusing on a different plane!



Quadrupole Matrix
(Thick lens)

- K_1 : Quadrupole strength [$1/\text{m}^2$]
- L_Q : Quadrupole length [m]
- k : Quadrupole integrated strength [$1/\text{m}$]

$$Q = \begin{pmatrix} \cos(\sqrt{K_1}L_Q) & \frac{\sin(\sqrt{K_1}L_Q)}{\sqrt{K_1}} \\ -\sqrt{K_1}\sin(\sqrt{K_1}L_Q) & \cos(\sqrt{K_1}L_Q) \end{pmatrix}$$

$$R = Q \times D$$

$$\Sigma = R\Sigma_0 R^T$$

$$\begin{pmatrix} \sigma_x^2 & \sigma_x \sigma'_x \\ \sigma_x \sigma'_x & \sigma'^2_x \end{pmatrix} = \varepsilon R \begin{pmatrix} \beta_0 & -\alpha_0 \\ -\alpha_0 & \gamma_0 \end{pmatrix} R^T$$

$$\sigma_x^2 = \dots = Ak^2 - 2ABk + C + AB^2$$

$$= A(k - B)^2 + C$$

Note: ABC ($\rightarrow$ and σ) depend on the elements of the matrix Σ (Σ_{11} , Σ_{12} , Σ_{21} , Σ_{22}).

The elements of the matrix Σ depends on Q (Q_{11} , Q_{12} , Q_{21} , Q_{22}), and D

See: AAD-LT-BM-AR-0138 (typo)

M.G. Minty and F. Zimmerman. Measurement and control of charged particle beams. Springer, 2003.

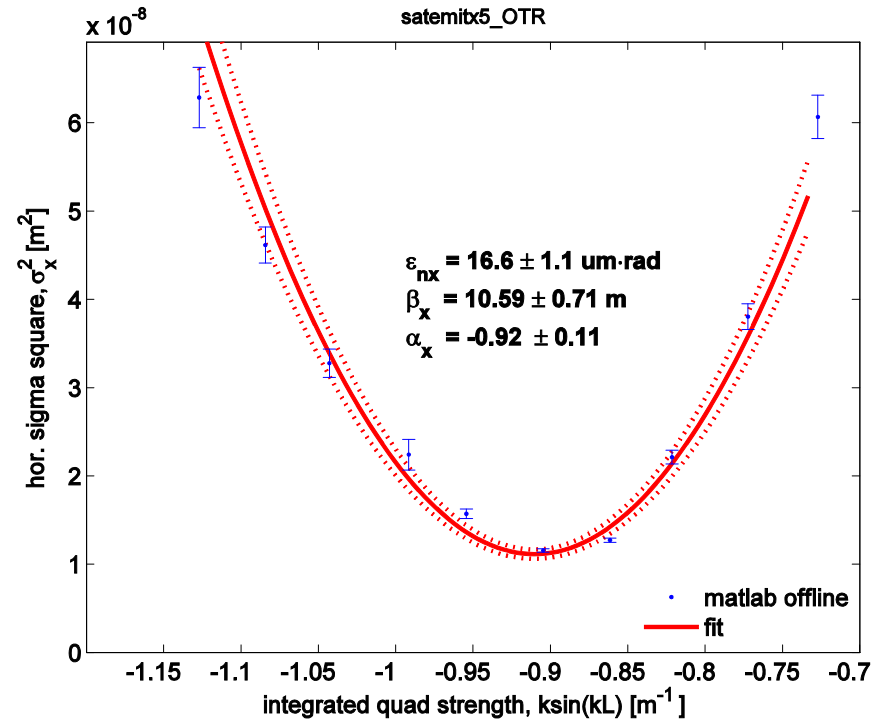
Data are taken for several quadrupole strengths and are fitted with a parabola
 $\sigma^2 = Ak^2 - 2ABk + C + AB^2$

$$\varepsilon = \frac{\sqrt{AC}}{\Sigma_{12}^2}$$

$$\beta_0 = \sqrt{\frac{A}{C}}$$

$$\alpha_0 = \sqrt{\frac{A}{C}} \left(\frac{1}{L} - B \right)$$

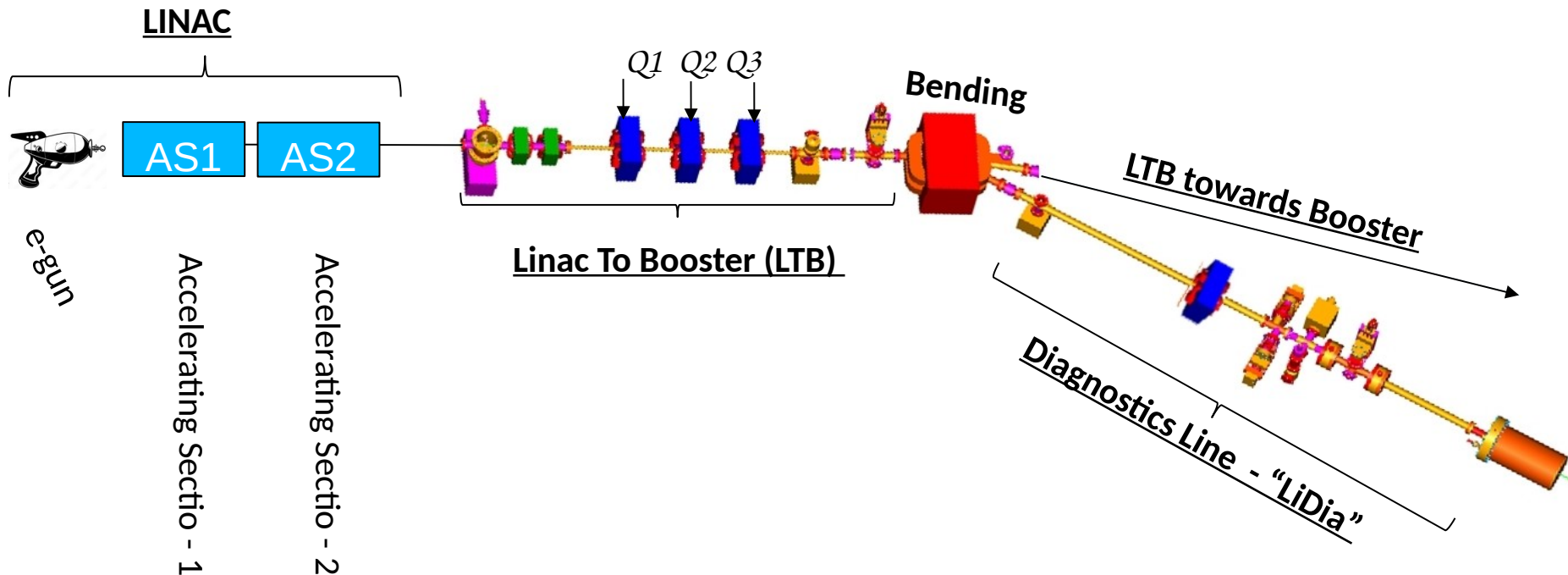
Twiss parameters are obtained from the fit parameters (A, B, C)
 $\Sigma_{12} = L$ (think lens approx.)



At ALBA, electrons are:

- generated at the e- gun of the ALBA Linac (90keV)
- accelerated in the two Accelerating Sections (AS1 & AS2) up to 125MeV (max)
- transferred to:
 - Booster through the Linac-to-Booster transfer line
 - Diagnostics Line (Lidia) for beam characterization

Linac set-up defines the beam **emittance**. In order to measure the beam emittance, we need to play with it equipment in both LTB and Lidia.

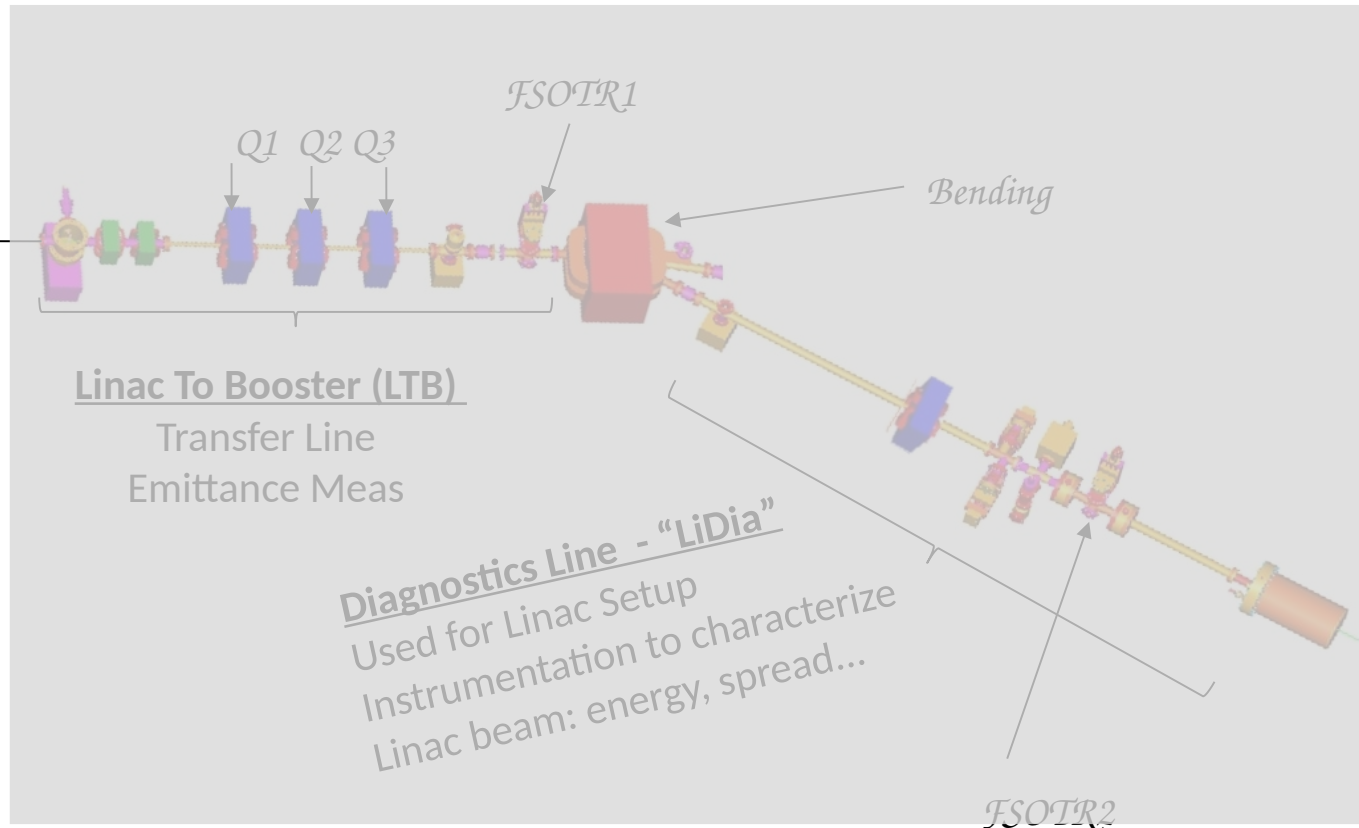
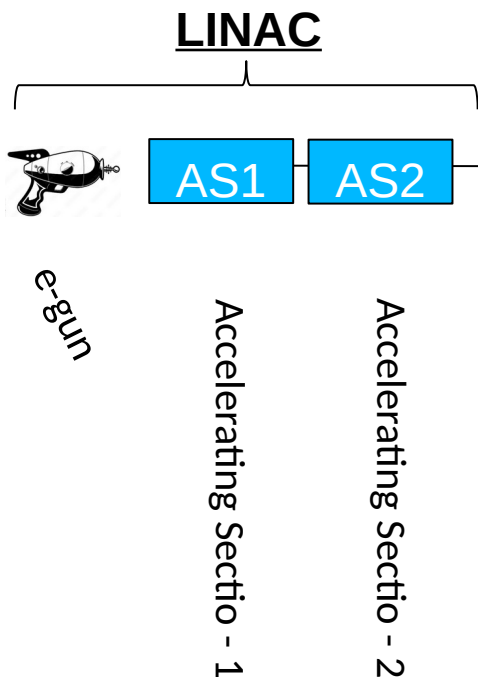


ALBA LINAC

LINAC

- Linac Length: 12m
- Thermoionic e-gun, heating up to $\sim 1000^\circ\text{C}$
- Two accelerating sections: AS1 & AS2
- Final beam energy: [60 – 125] MeV
- Emittance: $\sim 20 \text{ um}^*\text{rad}$ (hor and ver)

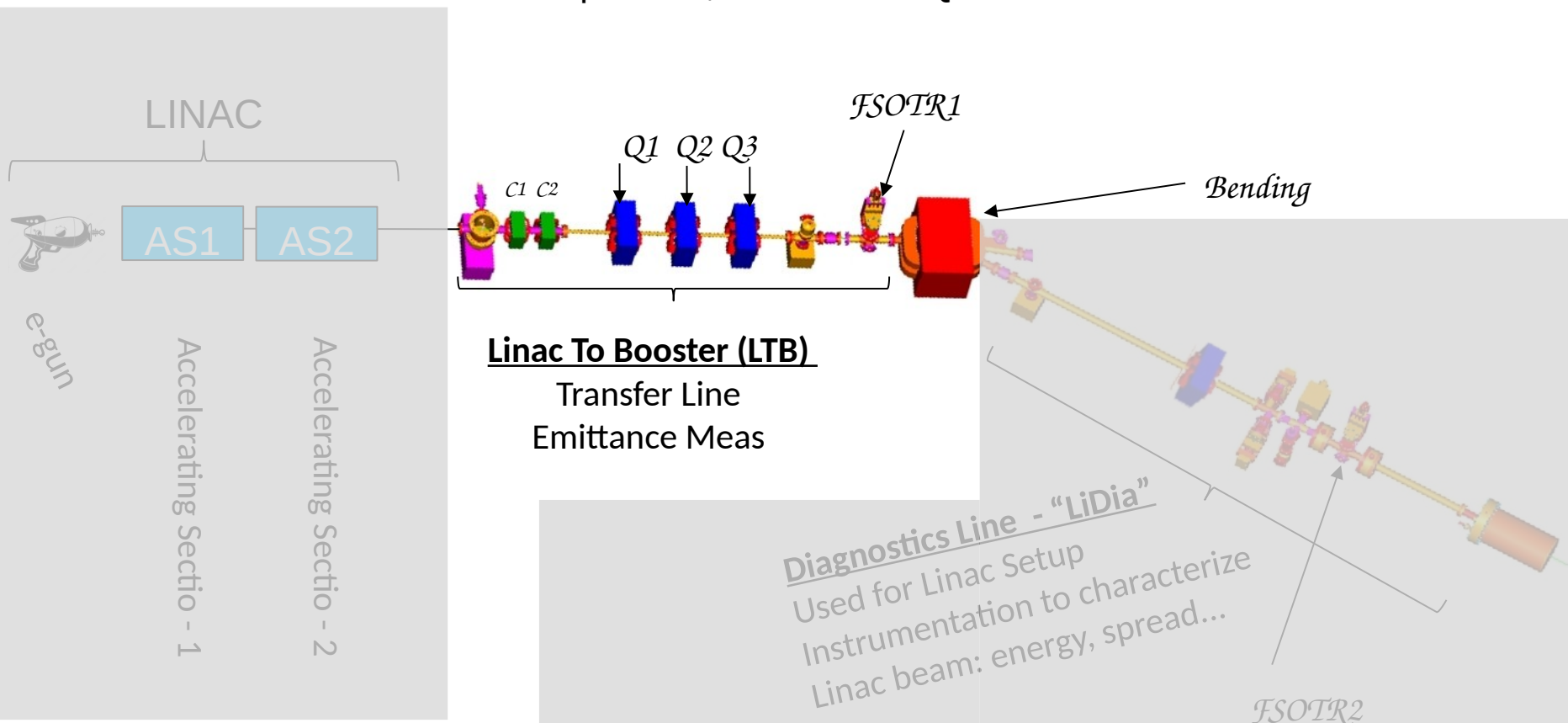
In this practice, we will only switch on/off the Linac or AS1/AS2



LTB

- Linac – To – Booster (LTB) transfer line
- Equipped with magnets to guide beam:
 - Dipoles, Quads (Q1, Q2, Q3...) , Correctors (C1, C2...) ...
- Equipped with instrumentation to diagnose the beam:
 - FSOTR1, FSOTR2, BPM, FCT...

In this practice, we will scan Q1 and measure beam size at FSOTR1

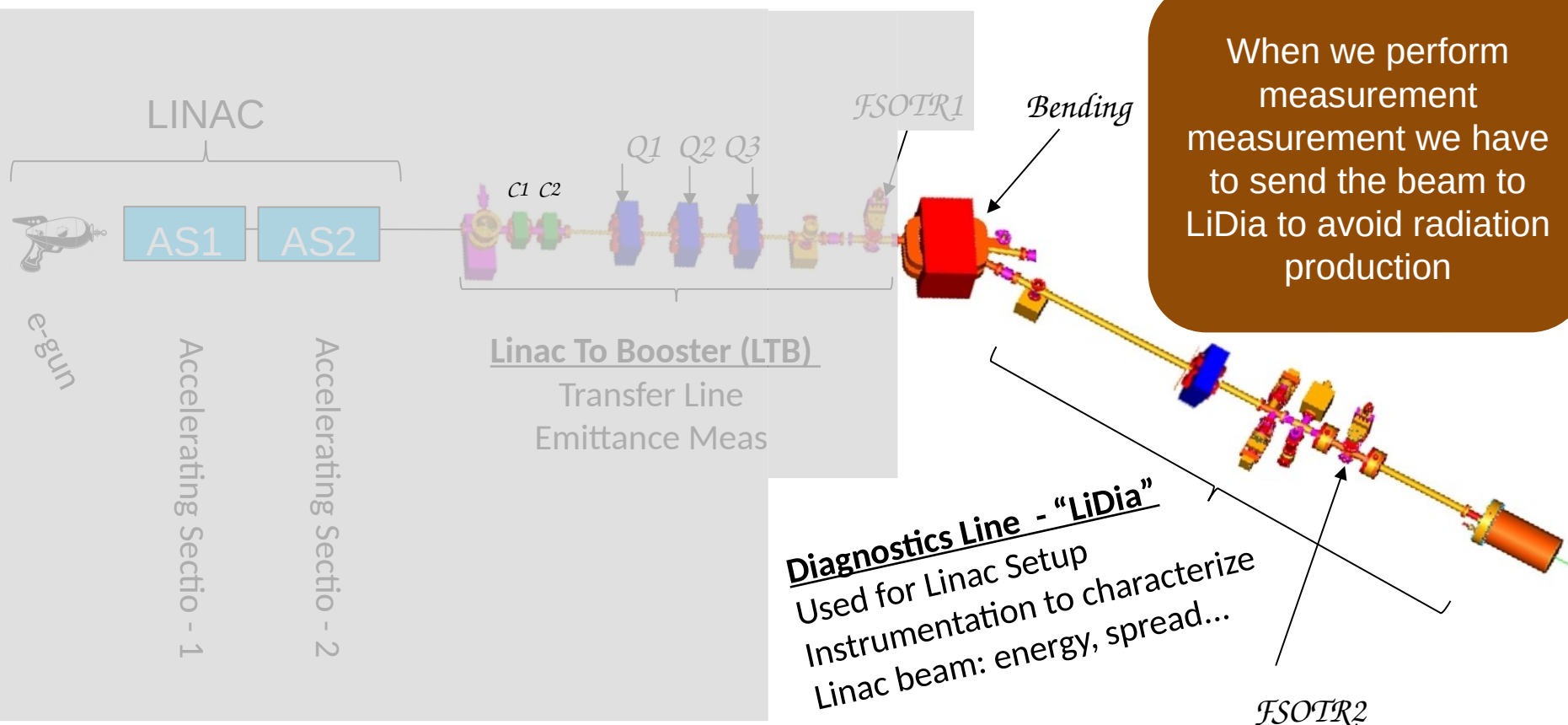


Diagnostic Line (LiDia)

LiDia

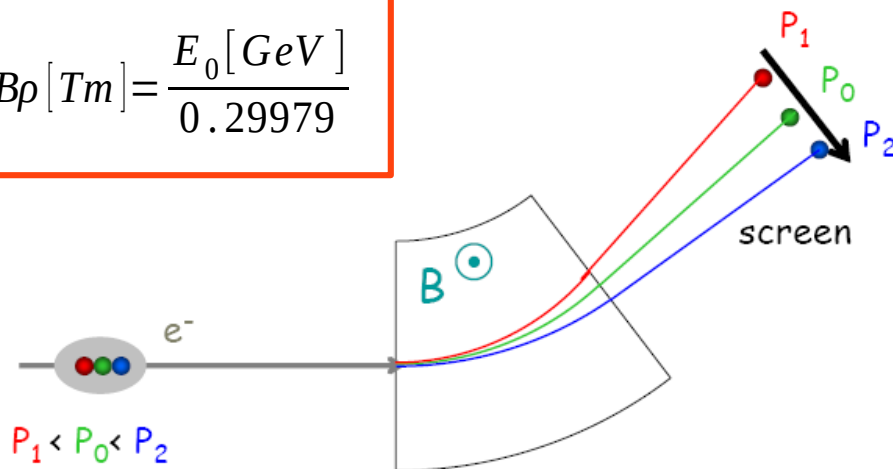
- Depending on the **dipole** intensity, beam can be sent to Booster or to LiDia for beam characterization and Linac set-up
- Equipped with one quad and diagnostics equipment
- Used to measure Linac energy and Energy Spread

In this practice, we will scan Dipole intensity and check at FSOTR_L



We use the bending magnet to deviate the beam to the LiDia and measure the beam energy

$$B\rho [Tm] = \frac{E_0 [GeV]}{0.29979}$$



Suppose to have an electron beam of **~80 MeV** and you have to deviate it of **30 deg** to send it to the LiDia. The bending magnet is **0.3 m** long. Which magnetic field we should we apply?

$$B = 0.4 \text{ T}$$

$$I = 111 \text{ A}$$

We change the magnetic field by varying the current in the bending magnet coils.

$$B = aI$$

$$a = 0.004 \text{ T/A}$$

What current shall we apply?

Adjusting the current we can center the beam in the screen and obtain the beam energy.

According to calibration:

$$E_0 = A'I + A'_0$$

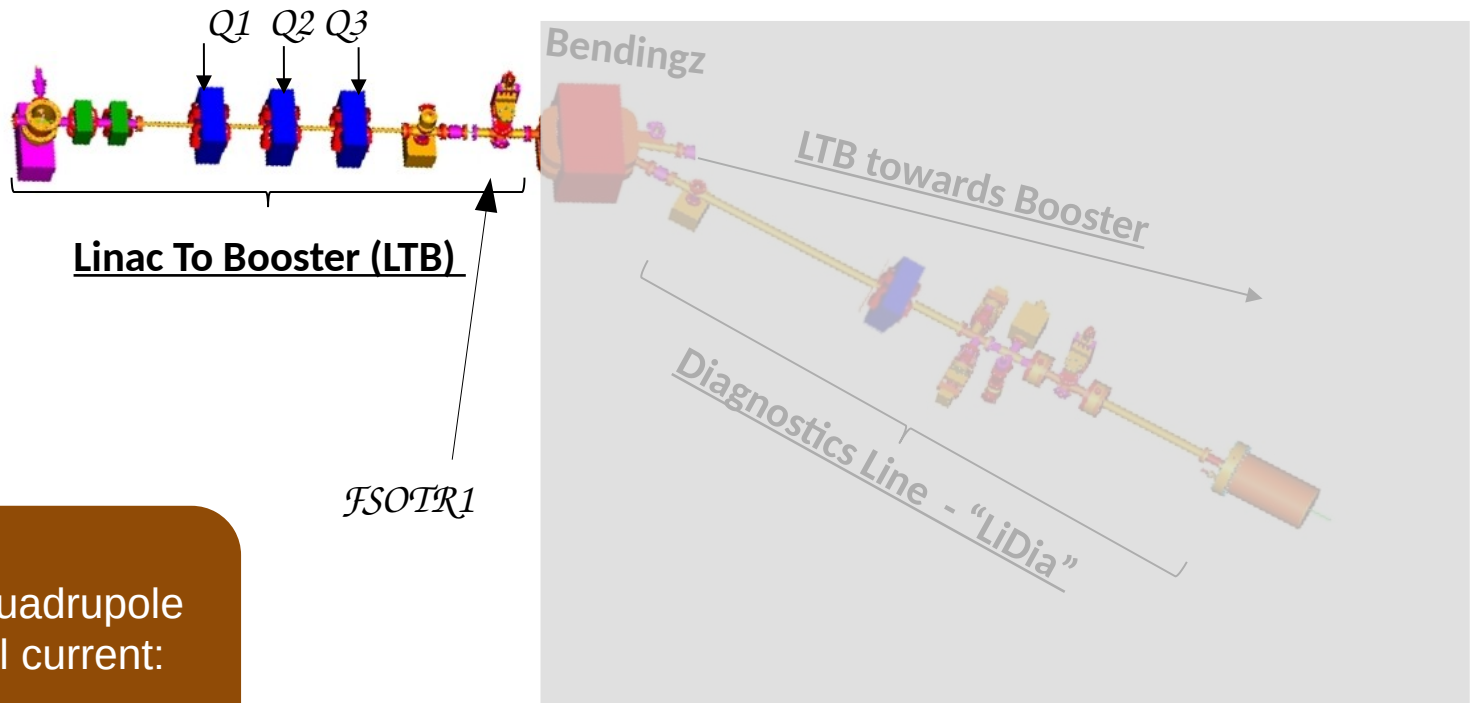
$$A' = 0.7058 \text{ MeV / A}$$

$$A'_0 = 0.2448 \text{ MeV}$$

Quadrupole Scan

For this measurement, we:

- Scan Q1 quadrupole strength by changing I in the Q1 coils
- Analyze the beam image obtained at with scintillator (YAG) at FSOTR1
- We will study the beam size for different focusing, and use the transport matrix to relate one with the other, knowing that the distance Q1-FSOTR1 = 1.235 m



Note:
Calibration quadrupole
strength / coil current:
 $k = I_{\text{quad}} * F/E$
 $F = 387.65 \text{ MeV/A/m}^2$